

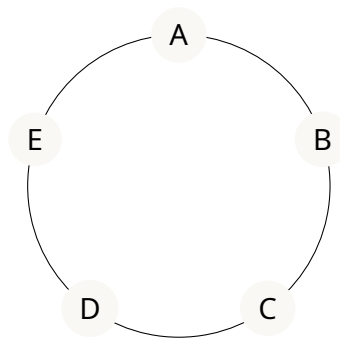


Assignment IV: Links

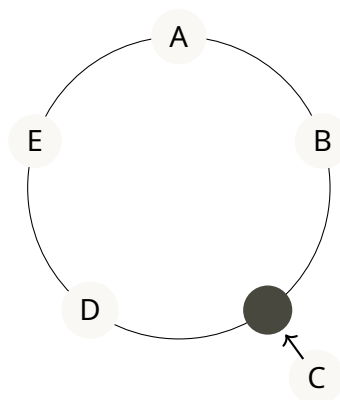
(Due date on Lea)

In this assignment, you will model a simple game played by people in a circle.

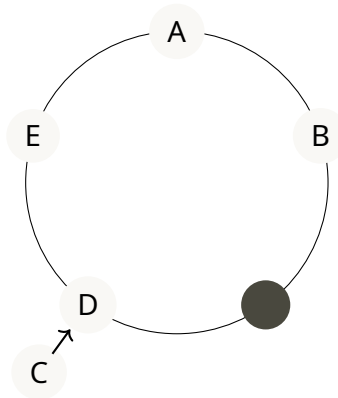
1. There are n people, labeled "A", "B", etc..., side-by-side forming a circle. For example, here is the circle for $n = 5$:



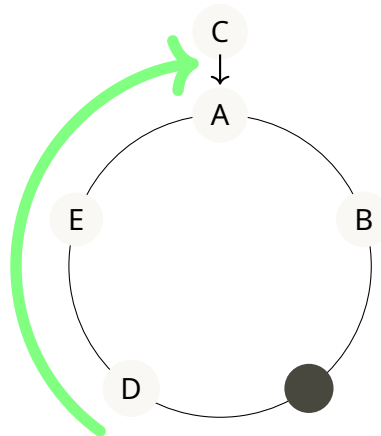
2. At the beginning of the game, the person in position m is *it*. The space they occupied will be the *exit* for the rest of the game. With the circle above and $m = 2$:



3. The person who is *it* moves one step clockwise to begin the game.

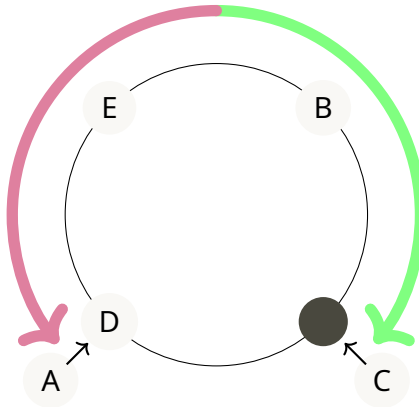


4. Starting from their current position in the circle, the person who is *it* counts clockwise around the circle to the o -th person and tags them. In our example, with $o = 3$:



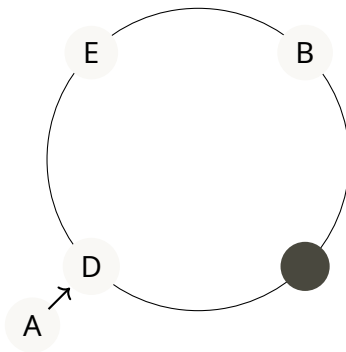
The exit is never part of the counting, i.e.: is skipped.

5. Once tagged, both players, *it* and *tagged*, will race towards the exit. The person who is *it* moves clockwise and the person *tagged* moves counter-clockwise, both at the same speed. The first one to arrive at the exit will leave.



If it's a tie, *tagged* will leave and *it* will stay and advance one step clockwise to move off of the exit.

6. If the person who is *it* exits before the person *tagged*, then *tagged* is now *it*.



7. The game continues until the circle is empty. The last person who is *it* at the end is the loser.

Examples

Here are some examples.

(n=5,m=0,o=3)

A escaped!
D escaped!
E escaped!
B escaped!
C is the loser.

(n=5,m=3,o=3)

D escaped!
B escaped!
C escaped!
E escaped!
A is the loser.

(n=7,m=2,o=3)

F escaped!
E escaped!
D escaped!
C escaped!
B escaped!
G escaped!
A is the loser.

(n=7,m=0,o=9)

D escaped!
E escaped!
B escaped!
A escaped!
G escaped!
C escaped!
F is the loser.

Requirements

Your program *must* meet the following requirements:

1. Read the integers n , m and o from user input (`input()`). Assume valid inputs are given by the user, that is
 - $0 < n \leq 26$
 - $0 \leq m < n$
 - $0 < o$
2. Solve this problem by using a *doubly-linked* chain. Use the following definition of a link in your solution:

```
from __future__ import annotations
from typing import TypeVar, Optional

T = TypeVar("T")

class DoubleLink[T]:
    def __init__(self, element: T):
        self.element: T = element
        self.next: Optional[DoubleLink[T]] = None
        self.prev: Optional[DoubleLink[T]] = None
```

This class is defined in `main.py`.

3. Print the order of people who escape the circle as well as the loser at the end of the game. This should look like:

```
n> 5
m> 3
o> 3
C escaped!
A escaped!
B escaped!
D escaped!
E is the loser.
```

Submission

Push your solution to the github classroom repository.

Style

Your program should be clear and well commented. It must follow the “420-4P6 Style Checklist” on Lea